

Evaluation Techniques for Regression and Classification

This guide explains evaluation metrics from beginner to advanced level, with intuition, formulas, numerical examples, Python code, comparison tables, real-life examples, interview questions, common mistakes, quick revision notes, a cheat sheet, and practice questions.

1) Regression Evaluation Metrics

Regression is used when the target is a continuous value, such as house price, salary, sales, demand, or temperature.

We'll reuse this sample data in several examples:

- 'y_true = [100, 120, 130, 150]'
- 'y_pred = [110, 115, 128, 140]'

Errors:

- '100 - 110 = -10'
- '120 - 115 = 5'
- '130 - 128 = 2'
- '150 - 140 = 10'

1.1 MSE ? Mean Squared Error

Simple Hinglish intuition

MSE batata hai ki model average mein kitna squared error kar raha hai. Pehle actual aur predicted ka difference nikaalte hain, phir square karte hain, aur phir average lete hain. Square karne ki wajah se bade errors ko zyada punishment milti hai.

Formula

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Terms

- 'n': number of observations
- 'y_i': actual value
- ' $\hat{y}_i$ ': predicted value
- ' $(y_i - \hat{y}_i)$ ': prediction error

Step-by-step example

Squared errors:

- '(-10)² = 100'
- '(5)² = 25'

- $(2)^2 = 4$
- $(10)^2 = 100$

Sum = $100 + 25 + 4 + 100 = 229$

$$MSE = \frac{229}{4} = 57.25$$

sklearn code

```
“python
from sklearn.metrics import mean_squared_error
```

```
y_true = [100, 120, 130, 150]
y_pred = [110, 115, 128, 140]
```

```
mse = mean_squared_error(y_true, y_pred)
print(mse)
““
```

When to use

- When large errors should be penalized strongly
- When discussing optimization/loss functions

When not to use

- When business users need easy interpretation in original units
- When outliers dominate the data

1.2 RMSE ? Root Mean Squared Error

Simple Hinglish intuition

RMSE, MSE ka square root hota hai. Isliye result target variable ki same unit mein aata hai. Agar house price predict kar rahe ho, to RMSE bhi price units mein hoga.

Formula

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Step-by-step example

We already found MSE = 57.25

$$RMSE = \sqrt{57.25} \approx 7.57$$

sklearn code

```
“python
from sklearn.metrics import mean_squared_error
import numpy as np

rmse = np.sqrt(mean_squared_error(y_true, y_pred))
print(rmse)
“
```

When to use

- When you want error in original units
- When large mistakes matter

When not to use

- When outliers are too extreme
- When you need percentage-based reporting

1.3 MAE ? Mean Absolute Error

Simple Hinglish intuition

MAE batata hai ki model average mein kitna absolute error kar raha hai. Isme square nahi hota, sirf absolute difference liya jata hai, isliye outliers ka effect MSE/RMSE se kam hota hai.

Formula

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Step-by-step example

Absolute errors:

- $|-10| = 10$
- $|5| = 5$
- $|2| = 2$
- $|10| = 10$

Sum = '27'

$$\text{MAE} = \frac{27}{4} = 6.75$$

sklearn code

```
“python
from sklearn.metrics import mean_absolute_error
```

```
mae = mean_absolute_error(y_true, y_pred)
print(mae)
""
```

When to use

- When interpretability matters
- When you want average error directly
- When outliers should not dominate too much

When not to use

- When large errors must be punished heavily

1.4 R Score ? Coefficient of Determination

Simple Hinglish intuition

R batata hai ki model ne target variable ki variability ka kitna hissa explain kiya.

'1' means perfect fit, '0' means baseline mean predictor jaisa performance, and negative means baseline se bhi worse.

Formula

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Terms

- ' $\bar{y}$ ': mean of actual values
- numerator: unexplained variation (RSS)
- denominator: total variation (TSS)

Step-by-step example

Mean of actuals:

$$\bar{y} = \frac{100 + 120 + 130 + 150}{4} = 125$$

RSS = '229'

TSS:

- ' $(100 - 125)^2 = 625$ '
- ' $(120 - 125)^2 = 25$ '
- ' $(130 - 125)^2 = 25$ '
- ' $(150 - 125)^2 = 625$ '

TSS = '1300'

$$R^2 = 1 - \frac{229}{1300} \approx 0.824$$

sklearn code

```
python
from sklearn.metrics import r2_score
```

```
r2 = r2_score(y_true, y_pred)
print(r2)
```

When to use

- To explain variance captured by the model
- To compare regression models broadly

When not to use

- As the only metric
- When you care more about actual business error size

1.5 Adjusted R

Simple Hinglish intuition

Regular R extra features add karne par aksar badh jata hai, chahe feature useful ho ya na ho. Adjusted R useless features ko penalize karta hai.

Formula

$$\text{Adjusted } R^2 = 1 - \left(\frac{(1 - R^2)(n - 1)}{n - p - 1} \right)$$

Terms

- 'R': regular coefficient of determination
- 'n': number of observations
- 'p': number of predictors/features

Step-by-step example

Assume:

- 'R = 0.824'
- 'n = 10'
- 'p = 2'

$$\text{Adjusted } R^2 = 1 - \left(\frac{(1 - 0.824)(10 - 1)}{10 - 2 - 1} \right)$$

$$\begin{aligned} &= 1 - \left(\frac{0.176 \times 9}{7} \right) \\ &= 1 - 0.2263 \\ &= 0.7737 \end{aligned}$$

sklearn/manual code

```
“python
from sklearn.metrics import r2_score

r2 = r2_score(y_true, y_pred)
n = 4
p = 2
adjusted_r2 = 1 - ((1 - r2) * (n - 1) / (n - p - 1))
print(adjusted_r2)
“
```

When to use

- Multiple regression
- Feature selection comparison

When not to use

- When there is only one feature
- When practical error metrics matter more

1.6 MAPE ? Mean Absolute Percentage Error

Simple Hinglish intuition

MAPE batata hai ki model average mein kitne percent ka error kar raha hai. Business dashboards mein ye kaafi popular hota hai.

Formula

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Step-by-step example

- $|100 - 110| / 100 = 0.10 = 10\%$
- $|120 - 115| / 120 = 0.0417 = 4.17\%$
- $|130 - 128| / 130 = 0.0154 = 1.54\%$
- $|150 - 140| / 150 = 0.0667 = 6.67\%$

Average percentage error:

$$\text{MAPE} = \frac{10 + 4.17 + 1.54 + 6.67}{4} \approx 5.60\%$$

\]

sklearn code

“python

from sklearn.metrics import mean_absolute_percentage_error

mape = mean_absolute_percentage_error(y_true, y_pred) * 100

print(mape)

““

When to use

- Forecasting and business reporting
- When percentage interpretation is useful

When not to use

- When actual values can be zero or near zero

2) Regression Real-Life Examples

House Price Prediction

- MAE tells average price error directly
- RMSE punishes expensive large mistakes more
- R shows how much price variation is explained
- MAPE gives percentage-style error reporting

Salary Prediction

If a few salary predictions are wildly wrong, RMSE exposes that strongly. If you want a directly understandable average miss, MAE is often better.

3) Classification Evaluation Metrics

Classification is used when the output is a category, like spam/not spam, disease/no disease, fraud/not fraud.

We'll use this confusion matrix example:

- TP = 40
- TN = 45
- FP = 5
- FN = 10
- Total = 100

3.1 Confusion Matrix

Simple Hinglish intuition

Confusion matrix ek table hota hai jo dikhata hai model ne kaunse predictions sahi kiye aur kaunse galat.

Terms

- **TP (True Positive):** actual positive, predicted positive
- **TN (True Negative):** actual negative, predicted negative
- **FP (False Positive):** actual negative, predicted positive
- **FN (False Negative):** actual positive, predicted negative

Real-life intuition

- Spam detection: FP means a real email went to spam
- Disease detection: FN means a sick patient was missed

sklearn code

```
“python
from sklearn.metrics import confusion_matrix
```

```
y_true = [1, 1, 1, 1, 0, 0, 0, 0, 1, 0]
y_pred = [1, 1, 0, 1, 0, 0, 1, 0, 1, 0]
```

```
cm = confusion_matrix(y_true, y_pred)
print(cm)
“
```

Visualization code

```
“python
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_true, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Pred 0', 'Pred 1'],
            yticklabels=['Actual 0', 'Actual 1'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
“
```

3.2 Accuracy

Simple Hinglish intuition

Accuracy batata hai total predictions mein se kitni correct thi.

Formula

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Step-by-step example

$$\text{Accuracy} = \frac{40 + 45}{40 + 45 + 5 + 10} = \frac{85}{100} = 0.85 = 85\%$$

sklearn code

```
python
from sklearn.metrics import accuracy_score
```

```
acc = accuracy_score(y_true, y_pred)
print(acc)
```

When to use

- Balanced classes
- Simple overview metric

When not to use

- Imbalanced datasets
- Medical/fraud use cases where one error type is dangerous

3.3 Precision

Simple Hinglish intuition

Precision ka answer: model ne jitni baar positive bola, unmein se kitni baar woh actually positive tha?

Formula

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Step-by-step example

$$\text{Precision} = \frac{40}{40 + 5} = \frac{40}{45} \approx 0.8889 = 88.89\%$$

Real-life example

Spam detection mein high precision ka matlab jab model spam bole, to mostly woh sach mein spam ho.

sklearn code

```
“python
from sklearn.metrics import precision_score

precision = precision_score(y_true, y_pred)
print(precision)
“
```

When to use

- False positives are costly
- Spam filtering, legal alerts, risk flags

When not to use

- When missing positives is more dangerous than false alarms

3.4 Recall

Simple Hinglish intuition

Recall ka answer: actual positive cases mein se model ne kitne pakde?

Formula

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Step-by-step example

$$\text{Recall} = \frac{40}{40 + 10} = \frac{40}{50} = 0.80 = 80\%$$

Real-life example

Disease screening mein recall high hona chahiye, kyunki sick cases miss karna dangerous hota hai.

sklearn code

```
“python
from sklearn.metrics import recall_score

recall = recall_score(y_true, y_pred)
print(recall)
“
```

When to use

- When false negatives are costly

- Medical screening, fraud detection, security monitoring

When not to use

- When too many false alarms are unacceptable

3.5 F1-Score

Simple Hinglish intuition

F1-score precision aur recall ka balanced summary hai. Ye tab useful hai jab dono important hon.

Formula

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Alternative:

$$F1 = \frac{2TP}{2TP + FP + FN}$$

Step-by-step example

Using precision = '0.8889' and recall = '0.80':

$$F1 \approx 2 \cdot \frac{0.8889 \cdot 0.80}{0.8889 + 0.80} \approx 0.842 = 84.2\%$$

sklearn code

```
“python
from sklearn.metrics import f1_score
```

```
f1 = f1_score(y_true, y_pred)
print(f1)
“
```

When to use

- When both precision and recall matter
- When classes are imbalanced and you need one summary metric

When not to use

- When TN matters a lot and should not be ignored

3.6 ROC Curve and AUC

Simple Hinglish intuition

Many classifiers probability dete hain, sirf final class nahi. Threshold change karne se TP/FP trade-off change hota hai. ROC curve different thresholds par TPR vs FPR dikhati hai.

Formula pieces

$$\backslash[\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \backslash]$$

$$\backslash[\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \backslash]$$

ROC plots 'FPR' on x-axis and 'TPR' on y-axis.

AUC means area under the ROC curve.

- '1.0' = perfect
- '0.5' = random guessing

Small threshold example

Suppose predicted probabilities are:

- positives: '0.90', '0.80', '0.60'
- negatives: '0.70', '0.40', '0.30'

At threshold '0.5':

- TP = 3
- FP = 1
- FN = 0
- TN = 2

So:

$$\backslash[\text{TPR} = 3/(3+0) = 1 \backslash]$$

$$\backslash[\text{FPR} = 1/(1+2) = 0.333 \backslash]$$

Different thresholds give different points.

sklearn code

```
"""python
from sklearn.metrics import roc_curve, roc_auc_score
```

```
import matplotlib.pyplot as plt
```

```
y_true = [1, 1, 0, 1, 0, 0]
```

```
y_scores = [0.90, 0.80, 0.70, 0.60, 0.40, 0.30]
```

```
fpr, tpr, thresholds = roc_curve(y_true, y_scores)
```

```
auc = roc_auc_score(y_true, y_scores)
```

```
print(auc)
```

```
plt.plot(fpr, tpr, marker='o', label=f'AUC = {auc:.2f}')
```

```
plt.plot([0, 1], [0, 1], '--', color='gray')
```

```
plt.xlabel('False Positive Rate')
```

```
plt.ylabel('True Positive Rate')
```

```
plt.title('ROC Curve')
```

```
plt.legend()
```

```
plt.show()
```

```
""
```

```
##### When to use
```

- Comparing classifiers by ranking ability
- Evaluating across many thresholds

```
##### When not to use
```

- When the dataset is highly imbalanced and PR metrics are more informative

```
---
```

3.7 Log Loss

```
##### Simple Hinglish intuition
```

Log loss evaluate karta hai ki model ne sirf sahi class predict ki ya nahi, balki kitni confidence se predict ki. Wrong and confident prediction ko heavy punishment milti hai.

```
##### Formula
```

```
\[
LogLoss = -\frac{1}{n}\sum_{i=1}^n\left[y_i\log(p_i) + (1-y_i)\log(1-p_i)\right]
\]
```

```
##### Step-by-step example
```

Suppose:

Actual Predicted Prob(1)

--- ---

1 0.9

0 0.2

1 0.8

0 0.1

Per-row losses are approximately:

- $-\log(0.9) = 0.105$
- $-\log(0.8) = 0.223$
- $-\log(0.8) = 0.223$
- $-\log(0.9) = 0.105$

Average:

```
\[  
LogLoss = (0.105 + 0.223 + 0.223 + 0.105)/4 = 0.164  
\]
```

sklearn code

```
“python  
from sklearn.metrics import log_loss
```

```
y_true = [1, 0, 1, 0]  
y_prob = [0.9, 0.2, 0.8, 0.1]
```

```
loss = log_loss(y_true, y_prob)  
print(loss)  
““
```

When to use

- When probability quality matters
- When calibration/confidence matters

When not to use

- When business users only care about final labels

4) Classification Real-Life Examples

Spam Detection

- Precision important: genuine mail should not go to spam
- Recall also useful: spam should not slip into inbox
- F1 useful if you want balance

Disease Prediction

- Recall critical: sick patients should not be missed
- Confusion matrix is essential to inspect FN specifically
- Log loss useful if probabilities drive medical review

5) Comparison Tables

5.1 Regression Metrics Comparison

Metric	Better Direction	Outlier Sensitivity	Easy Interpretation	Best For
MSE	Lower	High	Medium	Penalizing large errors
RMSE	Lower	High	High	Error in original units
MAE	Lower	Lower	Very High	Simple average error
R	Higher	Indirect	High	Variance explained
Adjusted R	Higher	Indirect	Medium	Multi-feature model comparison
MAPE	Lower	Problem near zero actuals	High	Percentage error

5.2 Classification Metrics Comparison

Metric	Focus	Best Use	Main Limitation
Accuracy	Overall correctness	Balanced classes	Misleading on imbalance
Precision	Positive prediction quality	FP costly	Ignores missed positives
Recall	Capture of real positives	FN costly	Can increase FP
F1-score	Balance of precision and recall	Both matter	Ignores TN
Confusion Matrix	Full error breakdown	Deep diagnosis	Not a single score
ROC-AUC	Ranking across thresholds	Compare classifiers	Less informative under severe imbalance
Log Loss	Probability confidence	Probabilistic models	Harder to explain

5.3 Which Metric When?

Regression

- Large mistakes matter most ? RMSE / MSE
- Direct average error wanted ? MAE
- Percentage reporting needed ? MAPE
- Explained variance needed ? R
- Comparing models with different predictor counts ? Adjusted R

Classification

- Balanced data ? Accuracy
- FP costly ? Precision
- FN costly ? Recall
- Both FP and FN matter ? F1-score
- Threshold-free ranking ? ROC-AUC
- Probability quality matters ? Log Loss

6) Important Interview Questions

Concept-based

1. What is the difference between MAE and RMSE?
2. Can R be negative? Why?
3. Why do we need Adjusted R?
4. Why is accuracy misleading on imbalanced data?

5. Explain precision vs recall in one sentence each.
6. Why does F1-score use harmonic mean?
7. Why can log loss be more informative than accuracy?

Scenario-based

1. For cancer screening, which metric matters most and why?
2. For spam detection, when would you prioritize precision over recall?
3. In house price prediction, when would you report RMSE instead of MAE?
4. In demand forecasting dashboards, why is MAPE popular?
5. How would you compare two regression models with different numbers of features?

Trick questions with short answers

1. Does high accuracy always mean a good model? ? No.
2. Is higher recall always better? ? Not always, because FP may rise.
3. Can precision be 1 while recall is low? ? Yes.
4. Can recall be 1 while precision is low? ? Yes.
5. Does low log loss guarantee highest accuracy? ? No.

7) Common Mistakes and Misconceptions

Regression mistakes

- Looking only at R and ignoring MAE/RMSE
- Using MAPE when actual values are zero
- Treating MAE and RMSE as identical
- Ignoring Adjusted R in multi-feature comparison

Classification mistakes

- Using accuracy on imbalanced datasets
- Confusing precision and recall
- Assuming F1 is always the best metric
- Ignoring threshold tuning
- Using ROC-AUC blindly for rare-event problems

8) Quick Revision Notes

Regression

- MSE: squared average error
- RMSE: same-unit version of MSE
- MAE: average absolute error
- R: explained variance
- Adjusted R: penalized R for extra features
- MAPE: percentage error, avoid zeros

Classification

- Accuracy: overall correct predictions

- Precision: when model says positive, how often is it right?
- Recall: among real positives, how many did it catch?
- F1: balance of precision and recall
- Confusion Matrix: TP, TN, FP, FN breakdown
- ROC-AUC: ranking quality across thresholds
- Log Loss: confidence/probability quality

9) Cheat Sheet

Regression Cheat Sheet

- 'MSE = average((y - y_hat)^2)'
- 'RMSE = sqrt(MSE)'
- 'MAE = average(|y - y_hat|)'
- 'R = 1 - RSS/TSS'
- 'Adjusted R = 1 - ((1-R)(n-1)/(n-p-1))'
- 'MAPE = average(|(y - y_hat)/y|) * 100'

Classification Cheat Sheet

- 'Accuracy = (TP + TN) / Total'
- 'Precision = TP / (TP + FP)'
- 'Recall = TP / (TP + FN)'
- 'F1 = 2PR / (P + R)'
- 'TPR = TP / (TP + FN)'
- 'FPR = FP / (FP + TN)'
- 'LogLoss = -avg(y log p + (1-y) log(1-p))'

10) Practice Questions

Regression practice

1. For 'y_true = [50, 60, 70]' and 'y_pred = [55, 58, 68]', calculate MAE, MSE, and RMSE.
2. Can a model have high R and still poor MAE? Explain with intuition.
3. Why is MAPE unsuitable when actual values include zero?
4. In salary prediction, when would you prefer RMSE over MAE?

Classification practice

1. If TP = 30, TN = 50, FP = 10, FN = 10, calculate Accuracy, Precision, Recall, and F1-score.
2. In fraud detection, which is worse: FP or FN? Why?
3. Why can 99% accuracy still mean a bad model?
4. How is ROC-AUC different from accuracy?

11) Compact sklearn Reference

```

#### Regression metrics
"""python
from sklearn.metrics import (
    mean_squared_error,
    mean_absolute_error,
    r2_score,
    mean_absolute_percentage_error,
)
import numpy as np

y_true = [100, 120, 130, 150]
y_pred = [110, 115, 128, 140]

mse = mean_squared_error(y_true, y_pred)
rmse = np.sqrt(mse)
mae = mean_absolute_error(y_true, y_pred)
r2 = r2_score(y_true, y_pred)
mape = mean_absolute_percentage_error(y_true, y_pred) * 100

print('MSE :', mse)
print('RMSE:', rmse)
print('MAE :', mae)
print('R2 :', r2)
print('MAPE:', mape)
"""

```

```

#### Classification metrics
"""python
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    roc_auc_score,
    log_loss,
)

y_true = [1, 1, 1, 1, 0, 0, 0, 0, 1, 0]
y_pred = [1, 1, 0, 1, 0, 0, 1, 0, 1, 0]
y_prob = [0.95, 0.90, 0.40, 0.80, 0.10, 0.20, 0.60, 0.30, 0.85, 0.15]

print('Accuracy :', accuracy_score(y_true, y_pred))
print('Precision:', precision_score(y_true, y_pred))
print('Recall :', recall_score(y_true, y_pred))
print('F1 Score :', f1_score(y_true, y_pred))
print('Confusion Matrix:\n', confusion_matrix(y_true, y_pred))
print('ROC AUC :', roc_auc_score(y_true, y_prob))

```

```
print('Log Loss :', log_loss(y_true, y_prob))
```

```
""
```

```
---
```

12) Final 10-Line Summary

1. Regression predicts numbers; classification predicts classes.
2. MSE and RMSE punish large errors more strongly.
3. MAE gives average error directly and is easier to explain.
4. R measures how much variance is explained.
5. Adjusted R is better than plain R for comparing models with different feature counts.
6. MAPE gives percentage error but fails near zero actual values.
7. Accuracy is useful mainly when classes are balanced.
8. Precision is critical when false positives are costly.
9. Recall is critical when false negatives are dangerous.
10. F1 balances precision and recall; ROC-AUC measures ranking; Log Loss measures probability quality.

APPENDIX: INTERVIEW Q&A

Evaluation Metrics Interview Q&A

Concept-Based Q&A

1. What is the difference between MAE and RMSE?

****Answer:**** MAE uses absolute errors, so each error contributes linearly. RMSE squares errors before averaging, so larger errors receive more punishment. Use MAE for simple average error and RMSE when large misses matter more.

2. Can R be negative?

****Answer:**** Yes. R becomes negative when the model performs worse than simply predicting the mean of the target for every observation.

3. Why do we need Adjusted R?

****Answer:**** Because regular R often increases when more features are added, even if those features are useless. Adjusted R adds a penalty for unnecessary predictors.

4. Why can accuracy be misleading?

****Answer:**** In imbalanced data, a model can predict the majority class almost always and still get high accuracy while being practically useless.

5. Precision vs Recall in one line?

****Answer:**** Precision asks, "When the model predicts positive, how often is it correct?"
Recall asks, "Out of all actual positives, how many did the model catch?"

6. Why does F1 use harmonic mean?

Answer: Because harmonic mean drops sharply when either precision or recall is low, so it prevents one high value from hiding one weak value.

7. Why is log loss powerful?

Answer: Because it evaluates not only whether the class is correct, but also whether the predicted probability is sensible and well calibrated.

Scenario-Based Q&A

8. Which metric matters most in cancer screening?

Answer: Recall. Missing a truly sick patient is usually more dangerous than sending a healthy patient for additional testing.

9. Which metric matters most in spam detection?

Answer: Often precision, because genuine messages should not be wrongly filtered. In practice F1 is also useful if both spam leakage and false filtering matter.

10. When should you report RMSE in house price prediction?

Answer: When large prediction mistakes should be penalized more strongly and the stakeholder wants error expressed in actual currency units.

11. Why is MAPE popular in forecasting dashboards?

Answer: Because percentage errors are easy for business teams to interpret and compare across products or regions.

12. Which metric would you use to compare multiple regression models with different numbers of predictors?

Answer: Adjusted R, because it accounts for the number of predictors while evaluating explained variance.

Trick Q&A

13. Does high accuracy always mean a good model?

Answer: No. A highly imbalanced problem can show high accuracy even when the model fails to detect the minority class.

14. Is higher recall always better?

Answer: Not always. Recall can be increased by predicting more positives, but that may also sharply increase false positives.

15. Can precision be perfect while recall is poor?

Answer: Yes. If the model predicts positive only a few times and all of them are correct, precision can be 1 while many actual positives are missed.

16. Can recall be perfect while precision is poor?

Answer: Yes. If the model predicts nearly everything as positive, it can catch all actual positives but create many false positives.

17. Does low log loss guarantee the highest accuracy?

****Answer:**** No. Log loss and accuracy evaluate different things. Accuracy uses hard labels; log loss uses probability quality.

Short Revision Q&A

18. Which regression metric is easiest to explain to business users?

****Answer:**** MAE, because it directly says the average prediction is off by a certain number of units.

19. Which regression metric is most sensitive to outliers?

****Answer:**** MSE, and therefore RMSE as well.

20. Which classification metric should you inspect first before all others?

****Answer:**** The confusion matrix, because it reveals the exact pattern of errors.